Matteo Paolini

Mechanical engineer · mechanical design, simulation & robot learning

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Available from February 2027

EXPERIENCE

Research Intern — Physical AI for robotic demining · Royal Military Academy, Brussels — 2026 – present

- Humanitarian demining: developing a reinforcement-learning policy (MuJoCo, teacher–student PPO) that decides which objects to remove, how and in what order, to extract a PFM-1 landmine safely from clutter; next: simulated UR5 and sim-to-real.

Machinery Maintenance Assistant — seasonal · PM Macchine, Corropoli (IT) — Summers 2016 – 2020

- Maintenance of leather-goods machinery (Adler, ATOM, Galli, OMAC) with experienced technicians or independently: sewing-machine hook timing, pneumatic troubleshooting, hydraulic seals, faulty sensors and control boards. Occasional support ongoing.
- Designed and 3D-printed many custom fixtures; most recent: a fixture positioning trouser labels during sewing, in PLA with a high wall count for stiffness.

Assembly & Machine Calibration Intern · Co.Me.Ta. Srl, Montegranaro (IT) — 2019

Assembled frames and integrated sensors, motors and electromechanical parts into industrial machines; final functional calibration.

Assembly & Wiring Intern · Galli S.p.A., Vigevano (IT) — assembled and wired leather-goods machinery — 2018

EDUCATION

M.Sc. Mechanical Engineering — Mechatronics · UNIVPM, Ancona · Erasmus at ULB & VUB, Brussels (2025–26) — 2024 – Feb 2027

Robotics, control system design, industrial automation, machine learning, composite structures, manufacturing.

B.Sc. Mechanical Engineering · UNIVPM, Ancona — 2021 – 2024

Thesis: collision-free pick-and-place planning for the redundant 7-DOF TIAGo arm (MATLAB) — RRT parameter study and path shortcutting.

SELECTED PROJECTS

Dual-arm object handover · Robotics II, VUB · academic project — 2025 – 2026

- Two Franka Panda arms pass a cube mid-air in Drake: offline handover-pose search from the arms' workspace intersection, IK, OMPL only where collisions are likely, 13-state coordinator.

Soft tentacle gripper · VUB Brussels · team of 5, supervised by Prof. B. Vanderborght — 2025

- Owned the tentacle: chamber geometry in ANSYS with a 2nd-order **Ogden hyperelastic** silicone model; two-part 3D-printed casting mold with removable core.
- Modular gripper with individually replaceable tentacles, integrated by the team on a 2-DOF arm (belt reductions from a 2.40 N·m torque requirement, Arduino pneumatics); public build guide and video.

Apple Watch case — production study · UNIVPM · team of 2 — 2025

- Reconstructed the Al 7005-T6 case in Siemens NX with functional GD&T (±0.01 mm fits, Ø0.05 concentricity on the Ø3 H7 crown bore); full process route from laser blanking to tumbling.
- Sized, on paper, a line for a **1,000,000 parts/year** target (1 laser, 89 CNC centers, 1 tumbler) from MRR and cutting power; removed a milling bottleneck by scheduling the fleet over four periods; Taylor tool-life and cost-per-part model.

Re-testing a published bearing-fault classifier · Electrical machines & drives, UNIVPM · academic project — 2026

- Reconstructed a DAE + CNN method (Sensors, 2021) on Paderborn motor-current data: **46.9%** accuracy vs the reported 99.6%. Identified SELU output clipping as the dominant error in my reconstruction (25.4% of samples unreachable); removing it raised AUC from 0.60 to 0.77, other training changes had no effect.

Fault diagnosis on a multiphase plant · Smart Factories, UNIVPM · academic project — 2026

- Showed a global detector learns the operating point, not the fault; with leakage-free temporal splits a regime-conditioned Random Forest detects 502–515 of 523 faults with **0 false alarms** on 798 healthy samples.

Also: vacUUUm cordless vacuum (Siemens NX product design) · 1U CubeSat CFRP/aluminum structure (ANSYS ACP, in progress) · LPBF 316L metallography and Vickers testing · PLC elevator controller (GRAFCET).

SKILLS

Robotics & ML	Python · MuJoCo · Gymnasium · Stable-Baselines3 · Drake · OMPL · scikit-learn · pytest · Git
Mechanical	Siemens NX · SolidWorks · ANSYS Workbench / ACP · hyperelastic FEA · GD&T & drawings · DFM · mold design · process planning
Prototyping	FDM 3D printing · silicone casting · Arduino · MATLAB / Simulink
Languages	Italian (native) · English (C1)